CS & IT ENGINEERING

Discrete Mathematics

Graph Theory

Lecture No.- 03

By- Satish Sir

Recap of Previous Lecture

Topic Basic of Graph

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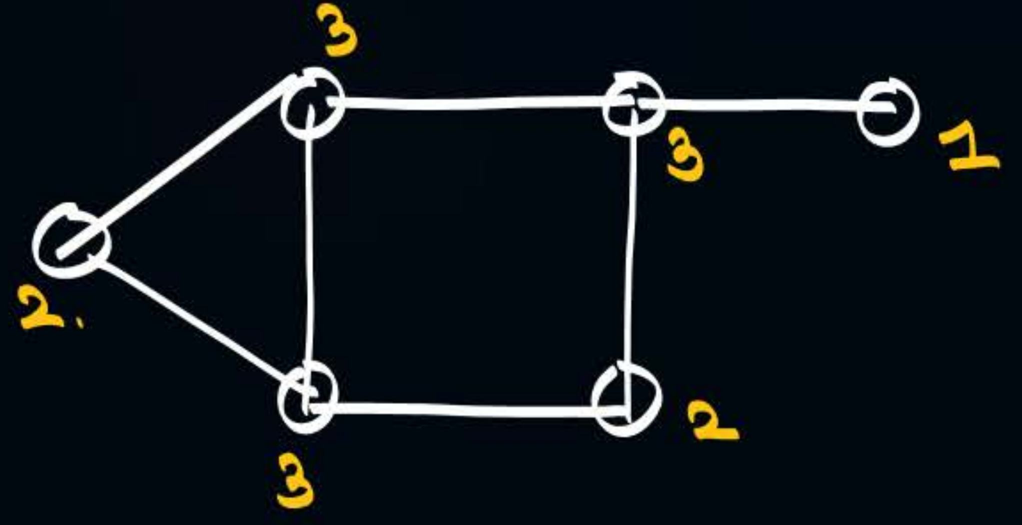
Topics to be Covered

Topic Graph Part-01

Topic : Graphs

Degree Sequence.:

writing degrees of all vertices either in increasing or in decreasing order.



→ 3, 3, 3, 2, 2, 1.

OR.

→ 1, 2, 2, 3, 3, 3.

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What will be total edges in the degree sequence 5, 2, 2, 2, 2, 1?

M1: $\sum d(v_i) = 2e$

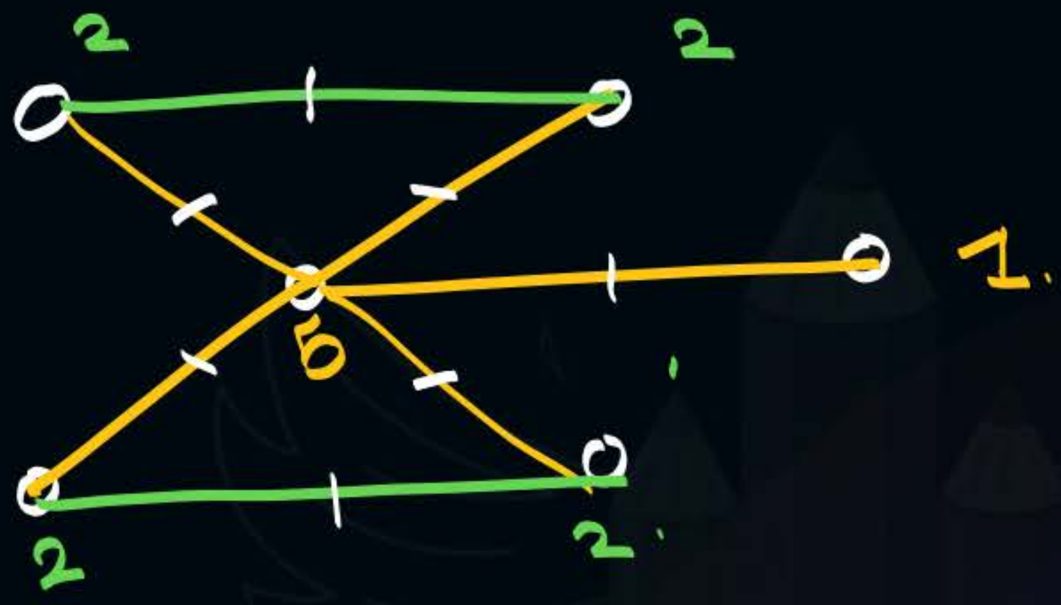
$$5+2+2+2+2+1 = 2e$$

$$14 = 2e$$

$$e = 7$$

M2: 5, 2, 2, 2, 2, 1

Total vertices = 6



$\boxed{e = 7}$

Q: What will be total edges in the degree sequence 3, 3, 3, 1?

M1:

$\sum d(v_i) = 2e$

$3+3+3+1 = 2e$

$10 = 2e$

$e = 5$

M2:

3, 3, 3, 1.

Total vertices = 4.

no simple Graph.

C is having 1 Degree

Requirement: 2.

2 edges.

Degree sequence.
- 5, 2, 2, 2, 2, 1 → Simple Graph. → Graphical sequence.
- 3, 3, 3, 1 → no Simple Graph. → not Graphical sequence.

Graphical sequence :

Degree sequence → Simple

Graphical ?

A) 5, 4, 3, 2, 1. ✗

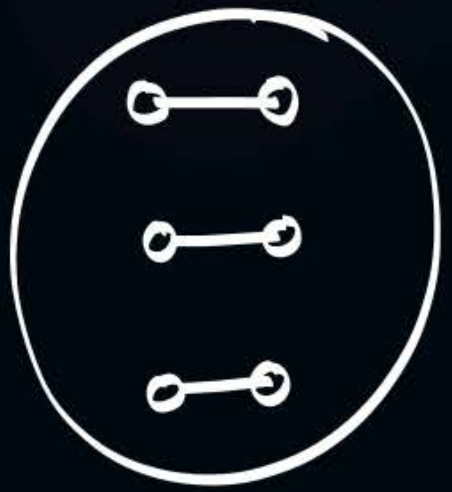
b) 4, 4, 3, 2, 1. ✗

c) 1, 1, 1, 1, 1, 1. ✓

d) 2, 2, 2, 2, 2, 2. ✓

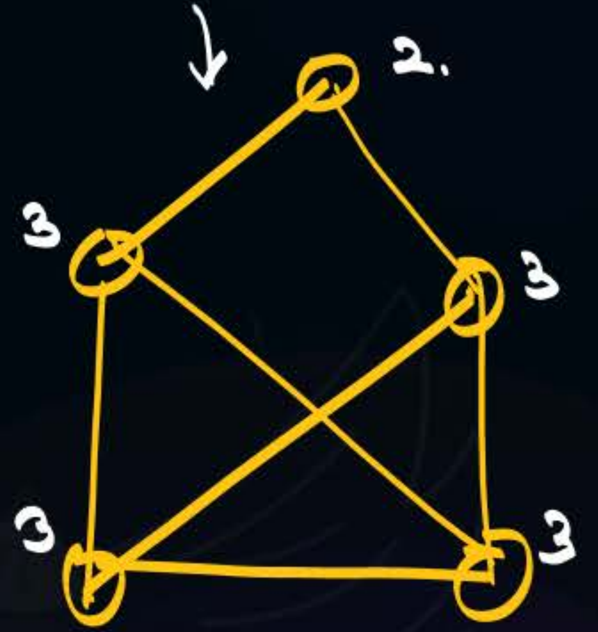
e) 3, 3, 3, 3, 2. ✓

c) 1, 1, 1, 1, 1, 1. ✓



d) 2, 2, 2, 2, 2, 2.

e) 3, 3, 3, 3, 2.



Graphical ?

A) 5, 4, 3, 2, 1.

1) Total vertices = 5

Thm3: $\Delta(G) \leq n-1$.

$\Delta(G) \leq 4$.

2) 5, 4, 3, 2, 1

no. of odd degree vertices must be even.

b) 4, 4, 3, 2, 1.

Thm3 ✓ $\Delta(G) \leq 4$. { Still it is not Graphical.

Thm2: ✓

$n-1, n-1$ ✓ . . . . . . 1.

4, 4, 3, 2, 1. Total vertices = 5

$n = 5$.

n vertices

$n-1, n-1 \ldots\ldots 1$

3, 3, 3, 1.

4, 4, 3, 2, 1 ×.

5, 5 .... 1 ×.

Total vertices = n.

$n-1, n-1 \ldots\ldots\ldots 1.$

then this sequence is not Graphical.

$n-1, n-1 \ldots\ldots\ldots$ may / may not be Graphical.

* $n-1, n-1 \ldots\ldots 1.$ not Graphical.

Total vertices = 3

$n = 3$

Degree sequence: 2, 2, 2.

(GATE)

Graphical ?

A) 7, 6, 5, 4, 4, 3, 2, 1.

B) 6, 6, 6, 6, 3, 3, 2, 2.

c) 7, 6, 6, 4, 4, 3, 2, 2.

d) 8, 7, 7, 6, 4, 2, 1, 1.

1) $\Delta(G) \leq n-1$.

2) Thm 2.

3) $n-1, n-1 \ldots\ldots 1$.

Havell-Hakimi Thm. :

$\hookrightarrow$ 20sec.

(GATE)

Graphical ?

A) 7, 6, 5, 4, 4, 3, 2, 1. ✓

B) 6, 6, 6, 6, 3, 3, 2, 2. ✗

c) 7, 6, 6, 4, 4, 3, 2, 2. ✓

d) 8, 7, 7, 6, 4, 2, 1, 1. ✗

~~7~~, 6, 5, 4, 4, 3, 2, 1. Graphical ✓

~~5~~, 4, 3, 3, 2, 1, 0

~~3~~, 2, 2, 1, 0, 0 (order)

1, 1, 0, 0, 0

1) Thm 2.

2) $\Delta(G) \leq n-1$.

3) $n-1, n-1 \ldots 1$.

4) Havell-Hakimi

Graphical?

A) 5, 3, 3, 3, 3, 2, 2, 2, 1.

B) 6, 3, 3, 3, 3, 2, 2, 2, 2, 1, 1.

c) 6, 5, 5, 4, 3, 2, 1.

d) 7, 5, 4, 4, 4, 3, 2, 1.

for which value of x.
the sequence is Graphical ?.

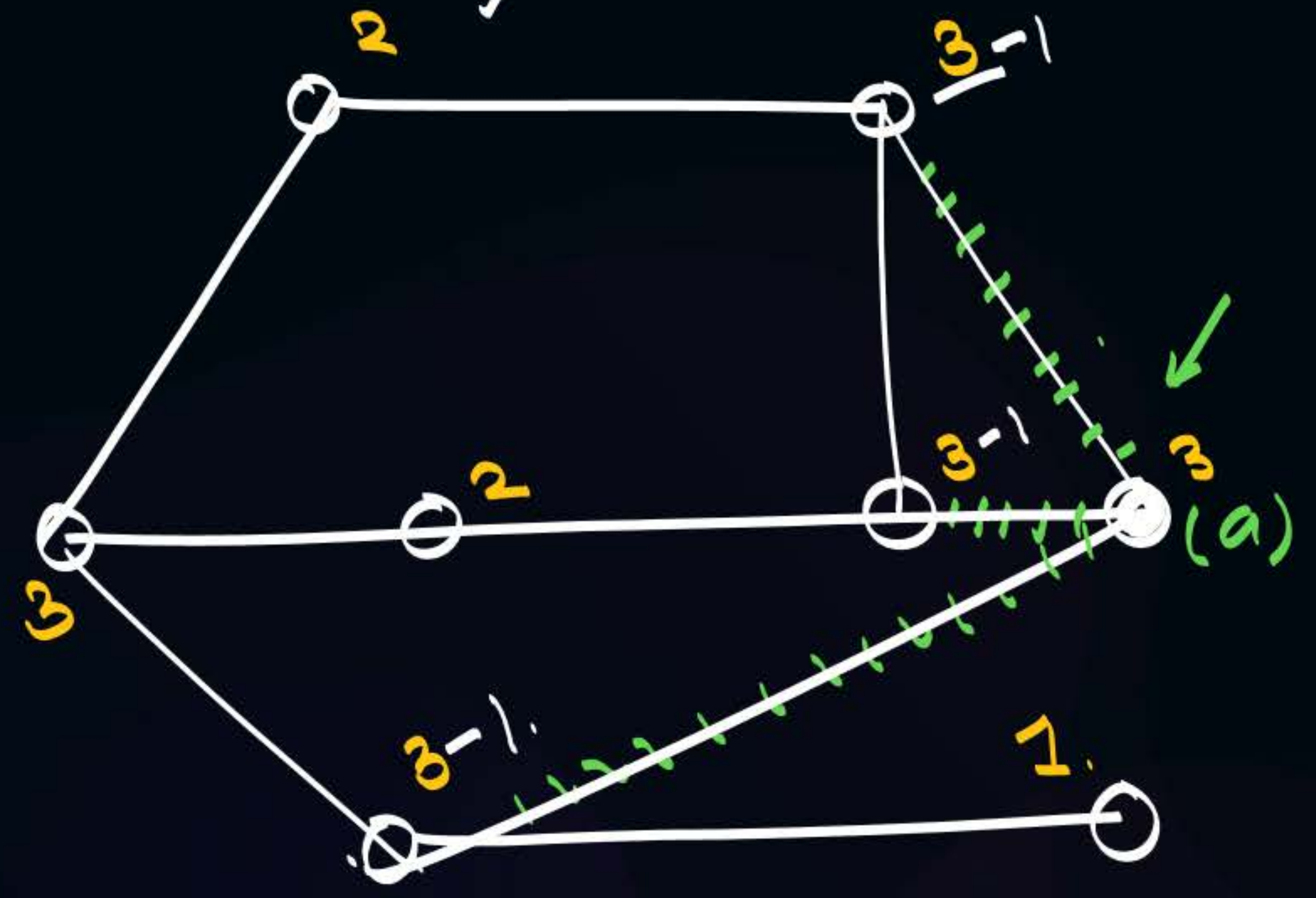
A) $x, 1, 2, 3, 5, 5$.

B) $7, 6, 5, 4, 3, 2, 1, x$.

C) $x, 7, 7, 5, 5, 4, 3, 2$.

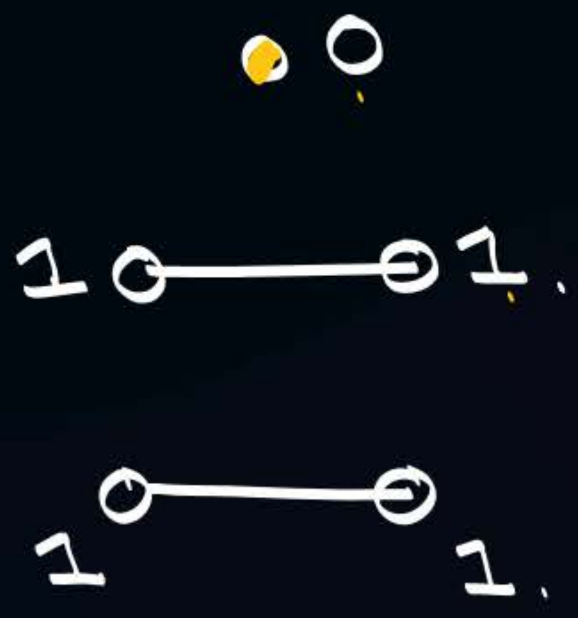
Havell-Hakimi :

→ ~~3~~, 3, 3, 3, 3, 2, 2, 1.

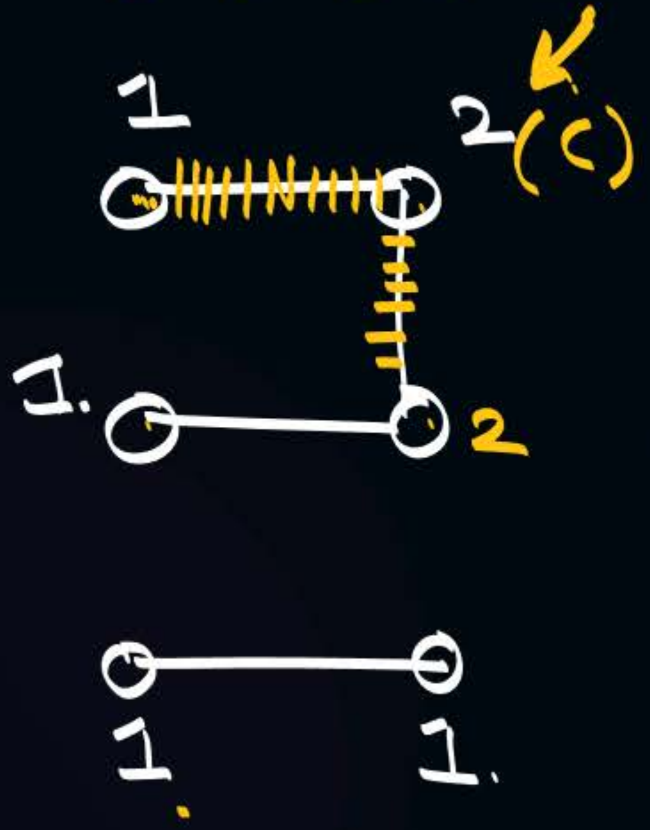


3, 2, 2, 2, 2, 2, 1.

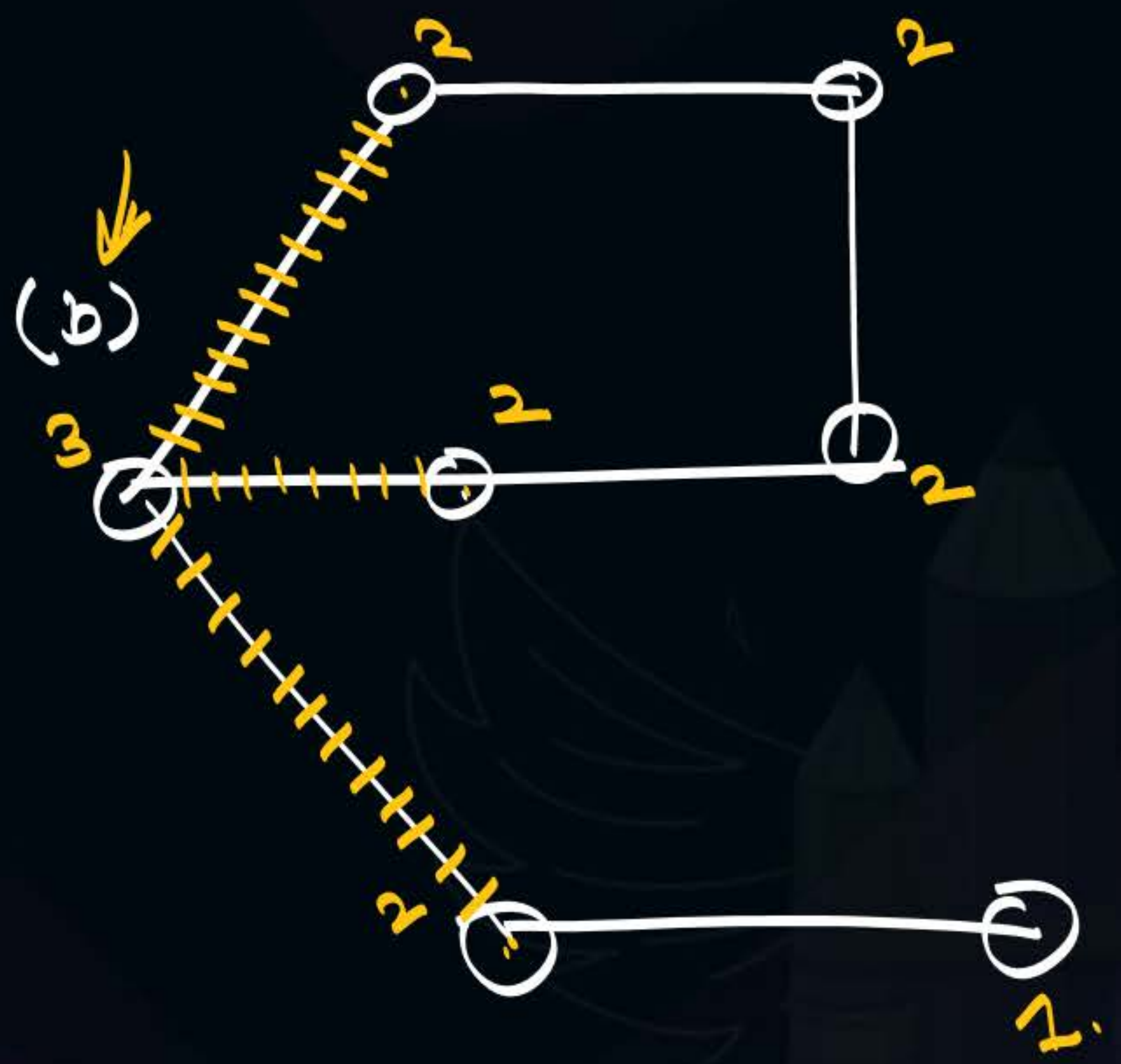
1, 1, 1, 1, 0



2, 1, 1, 1, 1.



3, 2, 2, 2, 2, 2, 1.



~~3~~, 3, 3, 3, 3, 2, 2, 1.

2, 2, 2, 3, 2, 2, 1

~~3~~, 2, 2, 2, 2, 2, 1. (ordering)

1, 1, 1 2, 2, 1.

~~2~~, 2, 1, 1, 1, 1.

1, 0, 1, 1, 1.

~~1~~, 1, 1, 1, 0 (ordering)

0, 1, 1, 0

1, 1, 0, 0 (ordering)

cut/count/mark.

→ Reduce by 1.

→ ordering.

a) $\cancel{5}, \dot{4}, \dot{3}, \dot{2}, \dot{1}.$

not Graphical

b) $\cancel{4}, \dot{4}, \dot{3}, \dot{2}, \dot{1}.$

$\cancel{3}, \dot{2}, \dot{1}, \dot{0}$ (order)

$1, 0, -1.$

not Graphical.

2 mins Summary

Topic One

Graph Part-01

Topic Two

Topic Three

Topic Four

Topic Five

THANK - YOU

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